

linearly from $-V$ to 0 .

20 C

By potential divider method, the p.d. across internal resistance is $E \frac{\left(\frac{r}{2}\right)}{R + \frac{r}{2}} = E \frac{r}{2R + r}$

$$\text{Fraction of Power} = \frac{P_r}{P_{total}} = \frac{IV_r}{IV_{total}} = \frac{V_r}{E} = \frac{r}{2R + r}$$

